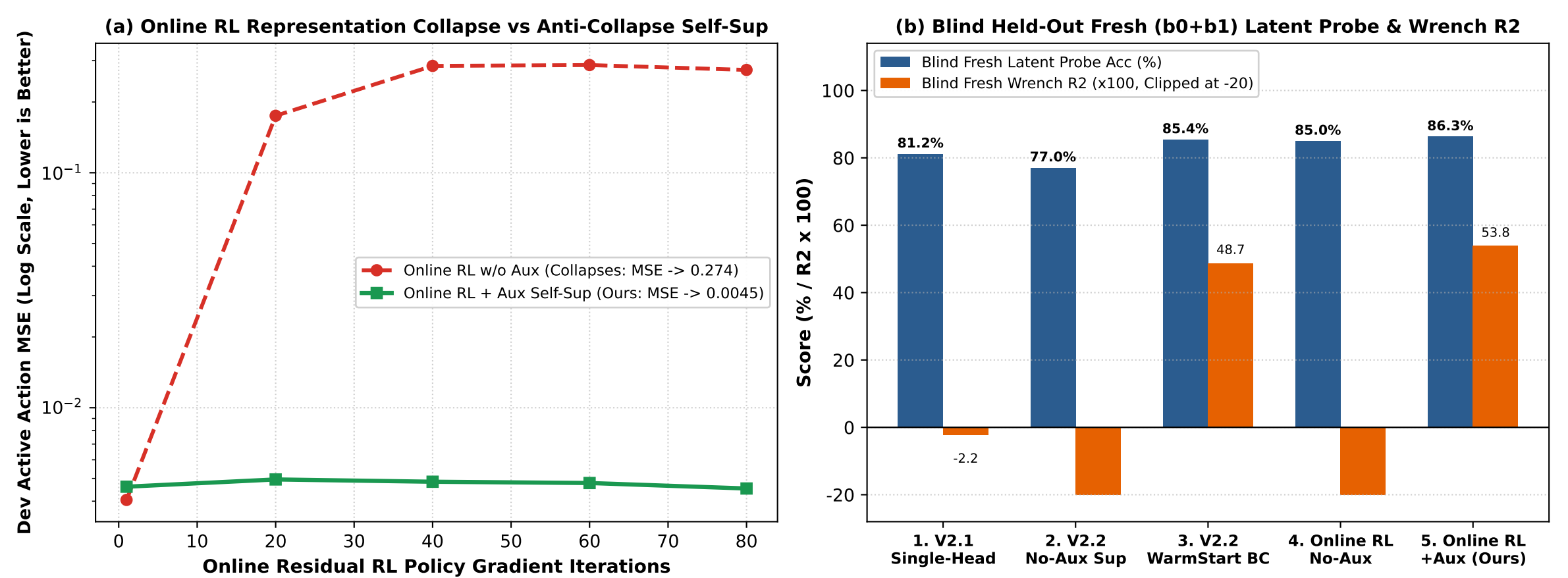
(a) Online RL Representation Collapse vs Anti-Collapse Self-Sup
Dev Active Action MSE (Log Scale, Lower is Better)
10^{-1}
10^{-2}
Online RL w/o Aux (Collapses: MSE -> 0.274)
Online RL + Aux Self-Sup (Ours: MSE -> 0.0045)
0
10
20
30
40
50
60
70
80
Online Residual RL Policy Gradient Iterations
(b) Blind Held-Out Fresh (b0+b1) Latent Probe & Wrench R2
Blind Fresh Latent Probe Acc (%)
Blind Fresh Wrench R2 (x100, Clipped at -20)
Score (% / R2 x 100)
100
80
60
40
20
0
-20
81.2%
-2.2
77.0%
85.4%
48.7
85.0%
86.3%
53.8
1. V2.1 Single-Head
2. V2.2 No-Aux Sup
3. V2.2 WarmStart BC
4. Online RL No-Aux
5. Online RL +Aux (Ours)